

# Label-Free, Bias-Free Classification of Difficult 4D-STEM Diffraction with a Diffraction-Tailored Self-Supervised Model

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## Abstract

Four-dimensional scanning transmission electron microscopy (4D-STEM) of beam-sensitive, low-dose, poorly-ordered materials produces diffraction data that are hard to segment into meaningful regions without labels, a forward model, or subjective choices such as the number of clusters or components. We introduce a self-supervised classifier purpose-built for nanobeam electron diffraction (NBED): a truncated convolutional DINO encoder trained with azimuthal ( $\theta$ -roll) augmentation for in-plane-rotation invariance, a radial-gated supervised-contrastive objective, and a one-dimensional radial clustering loss, with emergent cluster count and optional pairwise semi-supervision. The model needs no labels, no preset  $K$ , and no per-feature thresholds. On a sodium-phyllsilicate (NaPHI) flake it reproduces our previously published, expert-derived domain segmentation label-free and without the slow SAM + orientation-mapping pipeline (IoU 0.74, Dice 0.85;  $\sim 85\%$  of the indexed domain recovered, with near-identical region-average diffraction, Pearson  $r = 0.999$ ), and transfers to a new acquisition in milliseconds. A built-in interpretability protocol (representational probing, causal ablation, classical-baseline comparison, and ACOM cross-check) shows the clusters are governed by scattered intensity and low- $q$  two-dimensional diffraction structure, are rotation-invariant, and quantifies when the model is distinctive versus classical analysis: highly dis-

tinctive on hard low-dose molecular films (indomethacin, IMC) where classical decompositions cannot produce a coherent map, and convergent with classical analysis on well-ordered samples (NaPHI, EuInAs). The method is most valuable exactly where existing tools fail.

## Keywords

4D-STEM, nanobeam electron diffraction, self-supervised learning, unsupervised clustering, low-dose, interpretability, indomethacin

## Introduction

4D-STEM records a full diffraction pattern at every probe position, encoding local structure, orientation, strain and disorder across a field of view.<sup>1</sup> Turning that data cube into a meaningful map of distinct regions is a prerequisite for almost any downstream analysis. For robust, well-ordered crystals this is routine; for the beam-sensitive, low-dose, partially-ordered materials of greatest current interest—organic semiconductors, pharmaceutical polymorphs, layered and defective crystals—it is not. Patterns are sparse and noisy, domains are numerous, and diffuse scattering blurs the distinctions a human or a template would key on.

Unsupervised and self-supervised analysis is attractive because it needs no labels.<sup>2,3</sup> But the

standard automated routes each demand a subjective choice the microscopist must defend:  $k$ -means and density clustering require the number of clusters (or a density scale); non-negative matrix factorization (NMF) and principal component analysis (PCA) require the number of components;<sup>4</sup> supervised segmentation requires labels, which on hard data is prohibitive; and segmentation-model pipelines (e.g. SAM) plus orientation mapping are slow and parameterized. Just as importantly, even when a clustering succeeds, its *physical* meaning is rarely established—so the map cannot be trusted.

Here we address both problems. We present a self-supervised classifier engineered specifically for NBED, which produces a label-free, threshold-free first segmentation; we validate it against an independent, previously published expert result; and we pair it with an interpretability protocol that establishes what the clusters physically represent and when the method adds value over classical analysis.

## Results and Discussion

### A diffraction-tailored self-supervised model

Our classifier is a self-distillation (DINO) network,<sup>2</sup> but every major design choice is dictated by the physics of NBED rather than inherited from natural-image vision (Figure 1):

- **Convolutional, truncated encoder.** We use a truncated ResNet rather than a vision transformer. NBED patterns are small, sparse and low-count; the convolutional inductive bias and the reduced depth match the available signal, whereas transformers are data-hungry and over-parameterized for this regime.
- **Physics-chosen augmentations.** An azimuthal “ $\theta$ -roll” augmentation rotates the pattern in-plane during training, enforcing invariance to in-plane crystal rotation—a diffraction pattern rotated about the optic axis represents the *same*

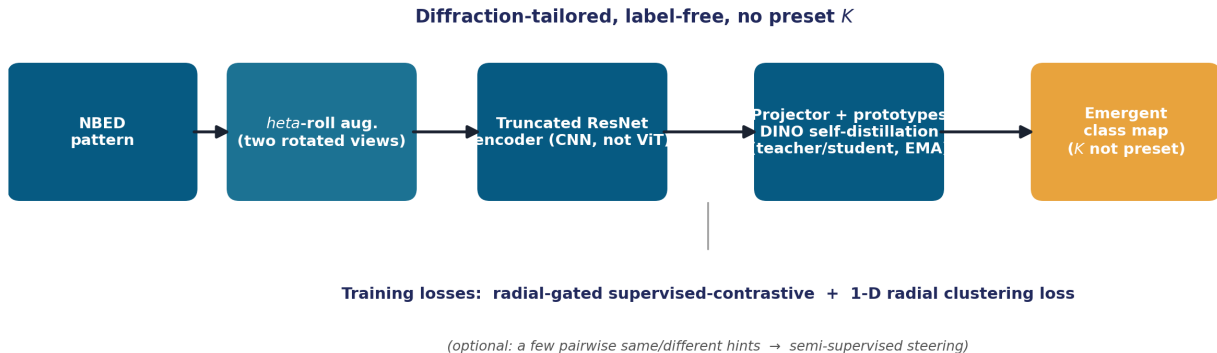
structure. Flips and colour-jitter, standard in vision, are disabled because they are meaningless or harmful for diffraction.

- **Radial-gated contrastive + 1-D loss.** A radial-gated supervised-contrastive term<sup>5</sup> and a one-dimensional radial clustering loss bias grouping toward the physically meaningful radial/ring signature.
- **Emergent  $K$  and optional semi-supervision.** We over-specify the number of prototypes; the model populates only those it needs, so the active cluster count emerges from the data rather than being preset. A small number of pairwise “same/different” hints can optionally steer the partition without reverting to full labelling.

The result is a single forward pass from raw cube to a class map, with no user-set  $K$ , components, or thresholds.

### Label-free concordance with expert segmentation

We first establish trust on a sample with an independent, published answer. In prior work on a NaPhi flake we segmented an oriented “Line” domain using a SAM segmentation plus orientation mapping with user-defined line parameters.<sup>6</sup> Running the present model on the same scan, with no labels and no line parameters, the union of two emergent classes reproduces that domain (Figure 2): IoU = 0.74, Dice = 0.85, with 85% of the SAM-indexed domain recovered. Moreover the region-average diffraction patterns are near-identical (beam-masked Pearson  $r = 0.999$  for both the Line domain and the rest, versus  $r \approx 0.5$  for Line-vs-rest), confirming the two segmentations are equivalent in diffraction space. The model achieves this without the SAM step, and a model trained on this flake transfers to a second, tilted acquisition in milliseconds—turning a slow, parameterized pipeline into an automatic one.



**Figure 1.** Model schematic. Each NBED pattern is augmented by an azimuthal  $\theta$ -roll (two rotated views) and passed through a truncated convolutional (ResNet) encoder; a DINO self-distillation head (teacher/student, EMA) with a radial-gated supervised-contrastive term and a 1-D radial clustering loss yields an emergent class map with no preset  $K$ . Optional pairwise hints allow semi-supervised steering.

## Resolving low-dose molecular films where classical methods fail

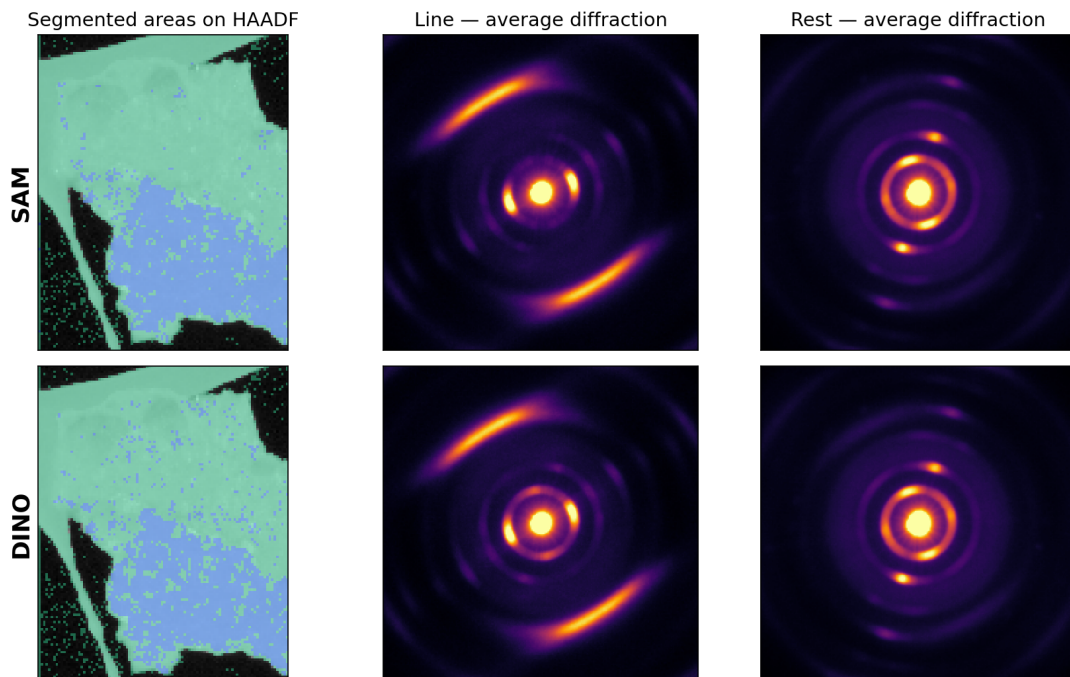
The payoff is on the hard target. On low-dose indomethacin (IMC) films the model produces coherent, fine-grained class maps, whereas classical automated methods do not. Clustering virtual dark-field, azimuthal-profile, PCA and NMF features of the *same* data into the same number of groups agrees only weakly with the DINO map (best adjusted Rand index, ARI  $\approx 0.11$ – $0.21$  across three IMC films; Figure 3). There is no external ground truth here—classical analysis simply cannot deliver a usable map—so the method is the only practical route, and its trustworthiness rests on the validation above and the interpretability below.

## What the clusters encode

To establish meaning we apply four complementary tests (Figure 4). *Representational probing*—5-fold ridge regression from the frozen embedding to candidate physical factors, with the correlation ratio  $\eta^2$ —shows the embedding most strongly encodes the total scattered intensity and crystallinity, and essentially ignores crystallographic orientation and Bragg-peak count. *Causal ablations* (re-inference on perturbed inputs, scored by ARI against the original map) disentangle the contributions. A

consistent post-mask intensity normalization (rescaling the kept, beam-masked region to constant) collapses the intensity-dominated samples (NaPHI, EuInAs  $\rightarrow$  1–2 classes) but leaves IMC with structure (7 classes, ARI 0.12): IMC depends on scattered intensity *and* low- $q$  2-D structure, not intensity alone. A complementary *per-ring* normalization—flattening the radial profile while preserving within-ring angular structure—collapses the molecular samples (NaPHI, IMC; i.e. radial/intensity-driven and rotation-invariant) but EuInAs retains ARI 0.29, showing it additionally uses angular/orientation structure, consistent with its zone-axis correlation (AMI  $\approx 0.30$ ). Azimuthal averaging collapses all samples (the full 2-D pattern is needed, not a 1-D profile), and masking the low- $q$  band is far more damaging than high- $q$ . *Classical-baseline comparison* quantifies novelty (above). *ACOM cross-check* confirms the partition is uncorrelated with indexed zone axis on the molecular samples (adjusted mutual information  $\approx 0$ ), consistent with the  $\theta$ -roll-imposed rotation invariance. Together: the clusters are driven by scattered intensity and low- $q$  2-D diffraction structure, are rotation-invariant, and their distinctiveness from classical analysis is material-dependent.

# Na007b — SAM (top) vs DINO (bottom): same domains, same diffraction



Column 1: Line (blue) and rest (green) areas overlaid on the virtual HAADF; vacuum = grayscale. Columns 2-3: average diffraction of each region (log, central 260 px). Line domains: IoU = 0.74, Dice = 0.85, containment(SAM in DINO) = 0.85. Beam-masked Pearson  $r$  of average patterns: SAM\_line vs DINO\_line = 0.999; SAM\_rest vs DINO\_rest = 0.999; Line vs rest = 0.48-0.54 (incl. cross terms) — the two methods isolate the same, diffraction-distinct, oriented domain.

**Figure 2.** NaPHI (Na007b): top = previously published SAM-derived segmentation, bottom = label-free DINO. Column 1: Line (blue) and rest (green) areas overlaid on the virtual HAADF (vacuum = grayscale). Columns 2–3: average diffraction of each region. The two methods isolate the same oriented “Line” domain (IoU 0.74, Dice 0.85) with near-identical region-average diffraction (beam-masked Pearson  $r = 0.999$  for both Line and rest), while Line and rest are diffraction-distinct ( $r = 0.48$ – $0.54$ ).

## Generality: crystalline, multi-phase, oriented data

The approach is not limited to molecular films. On a layered EuInAs sample with multiple phases, the model produces a sensible multi-domain map; classical methods here recover about half of it (ARI  $\approx 0.44$ ), i.e. the model is *least* distinctive on well-ordered material—it is most distinctive precisely on the hard molecular case. A three-phase ACOM analysis (EuInAs/In/As) further shows that, for this crystalline sample, the DINO classes additionally track crystal orientation (zone-axis AMI  $\approx 0.30$ ), a genuine secondary axis that the molecular films lack.

## A graphical interface for preparation, analysis and comparison

To make the method usable by experimentalists who do not program, we provide a graphical application built around three roles.

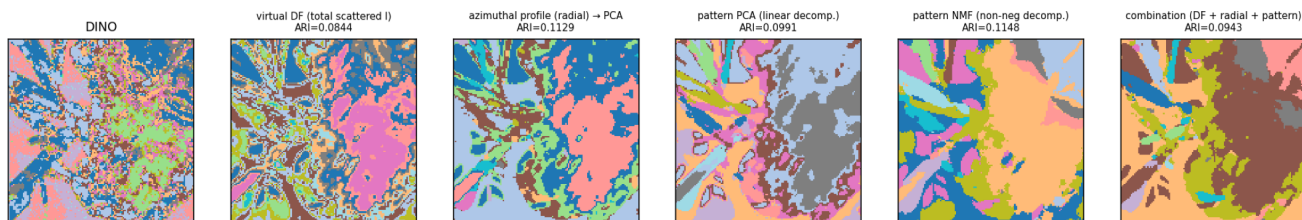
**(1) Data preparation.** It loads the common cube formats (.prz/.npy/.npz/.h5, including detector-linked masters), and offers the pre-processing that matters for hard data: intensity scaling (linear  $v_{\max}$  or logarithmic stretch), centre masking, real- or reciprocal-space Gaussian binning for low-dose denoising, region-of-interest selection, and multi-dataset handling—all non-destructive, writing new derived datasets.

**(2) Output analysis.** This is the core value. The class map is fully interactive: clicking a position shows its raw pattern; clicking a class shows the *class-average* diffraction; grains can

**All three IMC films — DINO map (far left of each row) vs classical clusterings of the same data**

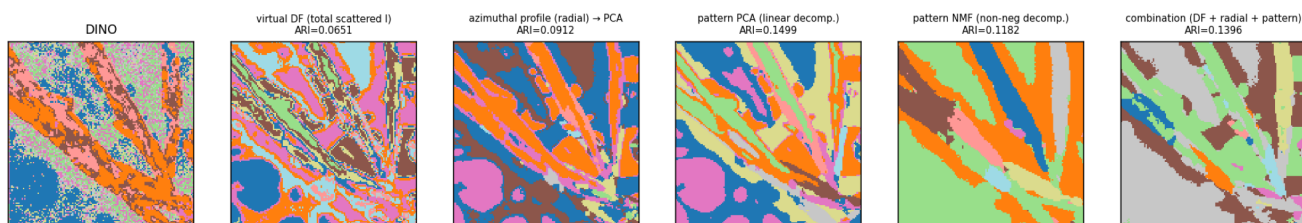
**IMC SI3 (K=12, classical best ARI 0.11)**

DINO vs classical clusterings (labels are arbitrary; shape match is what matters)



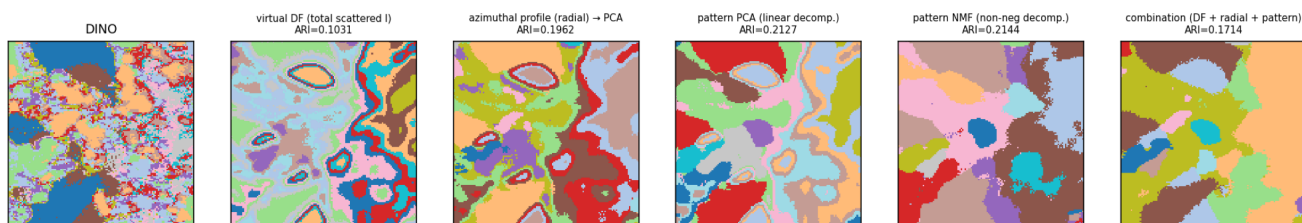
**IMC SI4 (K=9, classical best ARI 0.15)**

DINO vs classical clusterings (labels are arbitrary; shape match is what matters)

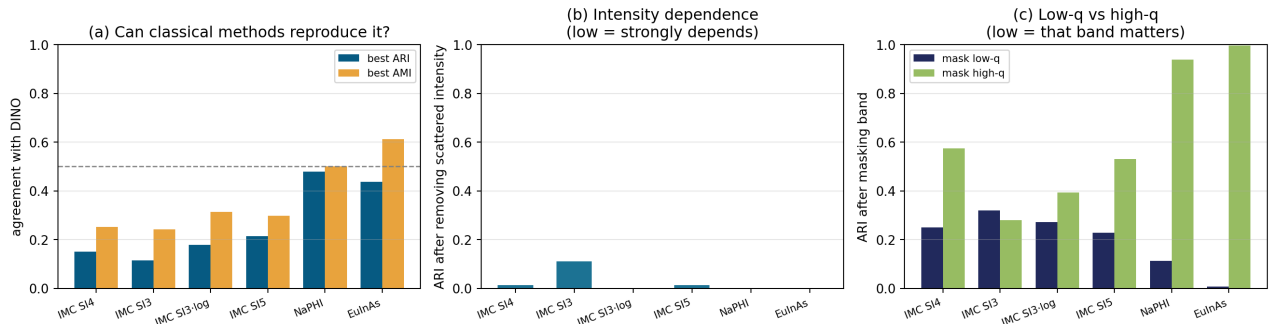


**IMC SI5 (K=13, classical best ARI 0.21)**

DINO vs classical clusterings (labels are arbitrary; shape match is what matters)



**Figure 3.** All three IMC films (SI3, SI4, SI5; rows): the DINO map (far left of each row) versus classical clusterings of the same data (virtual-DF, azimuthal-profile, PCA, NMF, combination). Colours are arbitrary; spatial shape is what matters. Across all three, classical methods recover only coarse regions and do not reproduce the DINO map (best ARI 0.11, 0.15, 0.21 for SI3/SI4/SI5).



**Figure 4.** Cross-sample interpretability (IMC SI3/SI4/SI5, NaPHI, EuInAs). (a) classical agreement with the DINO map (rises from distinctive IMC to substantially-captured NaPHI/EuInAs); (b) intensity dependence (low bar = strongly depends); (c) low- $q$  vs high- $q$  (low bar = that band matters—NaPHI/EuInAs depend almost entirely on low- $q$ ).

be averaged individually; and single-frame, per-grain and per-cluster averages can be stacked and compared side by side, with on-image  $q/d$ -spacing readout and gradient-attribution overlays. A one-click “Interpretation” tab runs the full battery above (probing, ablation, classical-baseline, ACOM cross-check) and writes a report—so the question “what do the clusters mean?” is answered inside the same tool that produced them.

**(3) Classical analysis, for comparison or standalone.** The same interface runs the classical methods on the same map—automated crystal orientation mapping (ACOM), NMF + clustering, crystallinity and spottiness mapping, and SAM-based segmentation—so a user can directly compare the self-supervised map against any of them, or use the classical tools on their own. The model is thus offered not as a black-box replacement but as the fast, unbiased first pass within a complete, comparison-ready toolkit.

## Conclusions

We have introduced a self-supervised, diffraction-tailored classifier that turns difficult low-dose 4D-STEM into a label-free, threshold-free, and physically interpretable domain map. It reproduces an independent published expert segmentation without labels or SAM, transfers across acquisitions in milliseconds, and—uniquely among the methods we compared—extracts a coherent map from low-dose molecu-

lar films that classical decompositions cannot. A built-in interpretability protocol establishes that the clusters encode scattered intensity and low- $q$  diffraction structure rather than orientation, and quantifies where the method adds value. Because it removes the subjective choices (labels,  $K$ , components, thresholds) that gate existing workflows while remaining auditable, it is well suited as a first-pass, high-throughput tool for beam-sensitive crystalline materials.

## Methods

**Model and training.** Truncated ResNet encoder; DINO self-distillation with teacher/student temperatures and EMA; azimuthal  $\theta$ -roll augmentation (student/teacher ranges), flips and colour-jitter disabled; radial-gated supervised-contrastive term and 1-D radial clustering loss; over-specified prototype count with emergent active  $K$ ; optional pairwise semi-supervision. **Data.** NBED cubes of IMC (low-dose molecular), NaPHI and EuInAs. **Interpretability.** Ridge-probe  $R^2$  and  $\eta^2$ ; causal input ablations re-inferred and scored by ARI; classical baselines (virtual-DF, azimuthal-PCA, pattern-PCA, NMF) clustered to the active  $K$  and scored by ARI/AMI; ACOM via py4DSTEM.<sup>7,8</sup> **Concordance.** IoU/Dice/containment of aligned, same-grid class and reference maps.

**Acknowledgement** Analyses were performed on frozen, previously validated model

checkpoints.

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